Project Plan: Embedded Traffic-Light Controller

Team: Lucas Alves Peralta, Neel, Rohan

1. Project scope and deliverables

Our project is an embedded traffic-light controller written in C++. The program simulates a traffic light at an intersection. It cycles through red, green, and yellow using set times for each light. It also has an input menu with a pedestrian button, so a user can request to cross and the light changes safely. The timing can be changed, and we will test edge cases like pressing the button during yellow or pressing it more than once.

At the end we will hand in:

• The C++ code in our GitHub repo, compiling with g++
• A state diagram of the light states and timing
• Test results logged in GitHub Issues
• One final PDF with our plan and documentation

2. Objectives

Team objectives

1. Have a working program that cycles red, green, and yellow correctly by Nov 13.
2. Have the pedestrian button and input menu working and tested with at least 5 cases by Nov 13.
3. Make the timing configurable, test the edge cases, and merge everything to main by Nov 20.
4. Track every task as a GitHub Issue with an owner and due date, and close it when it's done.
5. Have the final PDF and code in the repo by Dec 3.

Individual objectives

Person	My objectives	How I'll contribute
Lucas	Write the project scope and team objectives this iteration. Build the input menu and pedestrian button and test them with 5 cases by Nov 13. Help Rohan build the core light cycle by Nov 13. Take notes at every weekly meeting.	Project scope and goals, input and pedestrian button code, helping with the core logic, meeting notes
Neel	Set up the GitHub repo with a README, folder structure, and all teammates added this iteration.	Repo setup and Git help for the team
Rohan	Finish the project timeline and communication plan this iteration. Implement and test the core light cycle by Nov 13. Deliver the final formatted PDF by Nov 27.	Planning and scheduling, core logic, final document

3. Technologies and tools

Tool / language / framework	What we'll use it for	Why we picked it
C++	Main program	It's what we're learning in class
GitHub	Storing code and the plan	Required, and it tracks everyone's work
VS Code	Writing and editing the code	It is free and it is what we are most used to using
GitHub Issues	Tracking tasks and bugs	Keeps the to do list in the same place as the code
g++	Compiling and running the program	It matches the class environment

4. Project timeline

Task	Who	Deadline	Deliverable	Milestone?
Repo set up with README and folders	Neel	This iteration (done)	Repo link, all members added	Yes
Scope, objectives, tools sections	Lucas, Neel	This iteration (done)	Sections 1–3 drafted	
Design light states and timing (state diagram)	Rohan	Nov 6	Diagram in repo	
Core state machine (red/yellow/green cycle)	Rohan + Lucas	Nov 13	Compiles with g++, cycles correctly	Yes
Input menu / pedestrian button handling	Lucas	Nov 13	Working input, tested with 5 cases	
Timing configuration and edge cases	Neel	Nov 20	Tested code merged to main	
Full testing and bug fixes	Everyone	Nov 27	Test results logged in Issues	Yes
Final PDF assembled and formatted	Rohan	Nov 27	One clean PDF	
PDF uploaded and links submitted	Neel, everyone	Dec 3	PDF in repo, links submitted	Yes

5. Communication plan

Question	Our answer
How do we talk to each other?	Group chat (iMessage or Discord) for quick updates. Reply within 24 hours
When do we meet?	Once a week, e.g. after class or at a set evening time, plus extra if needed before deadlines
How do we track progress?	GitHub Issues: each task is an issue with an owner and due date, closed when merged.
What do we do if someone is stuck or behind?	Post in the chat as soon as you're blocked. If a task is 2+ days late, we reassign or pair up at the next meeting

6. Meeting summary

Meetings this iteration

Date	Who was there	What we decided
Oct 13	Everyone	Picked the traffic-light controller as our project. Split up the plan sections (Lucas: scope and objectives, Neel: tools and GitHub, Rohan: timeline and communication). Agreed on C++, VS Code, g++, and GitHub Issues, and set the meeting schedule below.

Upcoming meetings

Date	Who	Goal for this meeting
Nov 6	Everyone	State diagram finished. Start the core light cycle.
Nov 13	Everyone	Red/yellow/green cycle working. Input menu and pedestrian button tested with 5 cases.
Nov 20	Everyone	Timing settings and edge cases done and merged to main. Start full testing.
Nov 27	Everyone	All testing and bug fixes done, results logged in Issues. Final PDF drafted.

7. Contribution table

Person	What I completed this iteration	My responsibilities next iteration
Lucas	Wrote Section 1 (project scope and deliverables) and Section 2 (team objectives), helped split up the work, and took notes and wrote the meeting summary.	Build and test the input menu and pedestrian button with 5 cases, work with Rohan on the core red/yellow/green cycle, and keep taking meeting notes.
Neel	Created the GitHub repo, wrote the README and .gitignore, set up the src/ and docs/ folders, added teammates as collaborators, and wrote Section 3 (Technologies and tools)	Write and test the timer module, review teammates' code, and help with Git issues
Rohan	Worked on planning the project. Specifically completed the project timeline and communication plan	Design the state diagram and build the core light cycle with Lucas, then put together the final PDF